

SGT UNIVERSITY

School of Engineering and Technology

Department of CSE

DATABASE MANAGEMENT SYSTEMS LABORATORY

Experiment 4

SELECT Queries — DISTINCT, WHERE, BETWEEN, IN, ORDER BY

Field	Details
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Section	Section - C
Lab Date	09 September 2026
Dataset	employee — 10 sample rows × 8 attributes
Methods	SELECT, DISTINCT, WHERE, BETWEEN, IN, ORDER BY

1. Aim

To answer the lab questions below: create an employee table with the given attributes, insert 10 rows, and run the listed SELECT queries (DISTINCT, WHERE, BETWEEN, IN, ORDER BY).

2. Theory

2.1 SELECT

SELECT reads rows from a relation. SELECT * returns every column; naming columns returns only those attributes. A query without a WHERE clause returns every row that currently exists in the table.

2.2 DISTINCT

SELECT DISTINCT removes duplicate values from the result. Question 3 on the lab sheet asks for a “distant” department number; that is DISTINCT. SELECT department_no lists a department once per employee (question 4); SELECT DISTINCT department_no lists each department number only once.

2.3 WHERE

The WHERE clause keeps rows that satisfy a condition. Comparison operators such as > and <> filter numbers and strings. OR keeps a row if either predicate is true (city is Delhi or Mumbai). City values in this lab are stored in title case and matched with the same casing so the filters work in SQLite without extra case-conversion functions.

2.4 BETWEEN

BETWEEN low AND high is an inclusive range: the endpoints are part of the result. NOT BETWEEN is the complement — salaries strictly below 30000 or strictly above 50000. It is equivalent to salary < 30000 OR salary > 50000 when the range is 30000 to 50000.

2.5 IN

IN (...) tests membership in a list. city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore') keeps those four cities. NOT IN keeps every city that is not in the list — here Bangalore, Hyderabad, and Pune remain when Mumbai, Delhi, and Chennai are excluded.

2.6 ORDER BY

ORDER BY sorts the result. ASC (the default) lists names A→Z; DESC lists them Z→A. Sorting does not change stored rows; it only changes the order of the result set.

3. Schema

One table, employee, holds eight attributes. sr_no is the employee number (primary key). manager stores the manager's name, or NULL for the top manager.

Attribute	Type	Constraint	Role
sr_no	INTEGER	PRIMARY KEY	Employee number
employee_name	TEXT	NOT NULL	Employee name
job	TEXT	NOT NULL	Job title
manager	TEXT	NULL allowed	Manager name
hire_date	DATE	NOT NULL	Date of joining
salary	NUMERIC(10,2)	NOT NULL, CHECK > 0	Monthly salary
department_no	INTEGER	NOT NULL	Department number
city	TEXT	NOT NULL	Work city

4. Questions

The lab sheet numbered both CREATE and INSERT as 1; they are shown here as **1(a)** and **1(b)**. Question 3 wrote “distant”; the SQL keyword is DISTINCT. Employee number is stored as sr_no.

- 1(a).** Create a table employee with the following attributes:
 - sr no
 - employee name
 - job
 - manager
 - hire date
 - salary
 - department no
 - city
- 1(b).** Insert 10 rows in the above mentioned table.
- 2.** SELECT all the details from the employee table.
- 3.** Select only DISTINCT department number from the employee table.
The lab sheet wrote “distant” for this question.
- 4.** SELECT all department numbers from the employee table.
- 5.** SELECT employee name, salary WHERE salary is > 30000 rs from the employee table.
- 6.** SELECT employee name, hire date WHERE city is not Delhi from the employee table.
- 7.** SELECT employee name, hire date WHERE city is either Delhi or Mumbai from the employee table.
- 8.** SELECT employee number, employee name WHERE salary is BETWEEN 30000 rs and 50000 rs.
- 9.** SELECT employee number, employee name WHERE salary is NOT BETWEEN 30000 and 50000 from the employee table.
- 10.** SELECT all the details WHERE city is among the following: Delhi, Mumbai, Chennai, Bangalore.
- 11.** SELECT all the details WHERE city is not Mumbai, Delhi or Chennai from the employee table.
- 12.** SELECT all the details from the employee table and ORDER BY employee names in descending order.
- 13.** Display the list of employees in ascending order from the employee table.

5. Procedure

Drop employee if it already exists so the script can be re-run. Then answer each lab question in order. City values are stored in title case (Delhi, Mumbai, ...) and matched with that same casing. The SQL for each question:

Q. No.	SQL solution
1(a)	<pre>CREATE TABLE employee (sr_no INTEGER PRIMARY KEY, employee_name TEXT NOT NULL, job TEXT NOT NULL, manager TEXT, hire_date DATE NOT NULL, salary NUMERIC(10, 2) NOT NULL CHECK (salary > 0), department_no INTEGER NOT NULL, city TEXT NOT NULL);</pre>
1(b)	<pre>INSERT INTO employee (sr_no, employee_name, job, manager, hire_date, salary, department_no, city) VALUES (1, 'Rajesh Kumar', 'General Manager', NULL, '2018-03-12', 85000.00, 10, 'Delhi'), (2, 'Ananya Sharma', 'Software Engineer', 'Rajesh Kumar', '2021-07-01', 45000.00, 20, 'Bangalore'), (3, 'Rohan Mehta', 'Accountant', 'Rajesh Kumar', '2020-01-15', 28000.00, 30, 'Mumbai'), (4, 'Priya Nair', 'HR Executive', 'Rajesh Kumar', '2022-04-20', 32000.00, 40, 'Chennai'), (5, 'Vikram Singh', 'Sales Manager', 'Rajesh Kumar', '2019-11-08', 52000.00, 50, 'Hyderabad'), (6, 'Fatima Khan', 'Software Engineer', 'Ananya Sharma', '2023-02-14', 38000.00, 20, 'Bangalore'), (7, 'Arjun Patel', 'Clerk', 'Rohan Mehta', '2024-06-01', 18000.00, 30, 'Pune'), (8, 'Neha Gupta', 'Marketing Analyst', 'Rajesh Kumar', '2021-09-10', 35000.00, 60, 'Delhi'), (9, 'Karan Iyer', 'Database Admin', 'Ananya Sharma', '2020-12-05', 48000.00, 20, 'Mumbai'), (10, 'Meera Joshi', 'Receptionist', 'Priya Nair', '2023-08-22', 22000.00, 40, 'Chennai');</pre>
2	<pre>SELECT * FROM employee;</pre>
3	<pre>SELECT DISTINCT department_no FROM employee;</pre>
4	<pre>SELECT department_no FROM employee;</pre>
5	<pre>SELECT employee_name, salary FROM employee WHERE salary > 30000;</pre>
6	<pre>SELECT employee_name, hire_date FROM employee WHERE city <> 'Delhi';</pre>
7	<pre>SELECT employee_name, hire_date FROM employee WHERE city = 'Delhi' OR city = 'Mumbai';</pre>
8	<pre>SELECT sr_no, employee_name FROM employee WHERE salary BETWEEN 30000 AND 50000;</pre>
9	<pre>SELECT sr_no, employee_name FROM employee WHERE salary NOT BETWEEN 30000 AND 50000;</pre>
10	<pre>SELECT * FROM employee WHERE city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore');</pre>
11	<pre>SELECT * FROM employee WHERE city NOT IN ('Mumbai', 'Delhi', 'Chennai');</pre>
12	<pre>SELECT * FROM employee ORDER BY employee_name DESC;</pre>
13	<pre>SELECT * FROM employee ORDER BY employee_name ASC;</pre>

6. Source Code

SQLite script used in the lab compiler (09-09-2026/employee.sql). For the class SQL Server, run employee.sqlserver.sql.

```
-- =====
-- DBMS Lab · 09-09-2026
-- Employee table: CREATE, seed 10 rows, then SELECT queries
-- DISTINCT, WHERE, BETWEEN, IN, ORDER BY
-- SQLite (sql.js compiler / DB Browser / sqlite3)
-- For the class SQL Server, run employee.sqlserver.sql instead.
--
-- Lab questions (1(a)/1(b) = sheet's two "1." items; "distant" = DISTINCT)
-- 1(a). Create a table employee with the following attributes:
--      sr no, employee name, job, manager, hire date, salary,
--      department no, city
-- 1(b). Insert 10 rows in the above mentioned table
-- 2. SELECT all the details from the employee table
-- 3. Select only DISTINCT department number from employee table
--      (lab sheet wrote "distant")
-- 4. SELECT all department numbers from the employee table
-- 5. SELECT employee name, salary WHERE salary is > 30000 rs
--      from the employee table
-- 6. SELECT employee name, hire date WHERE city is not Delhi
--      from the employee table
-- 7. SELECT employee name, hire date WHERE city is either
--      Delhi or Mumbai from the employee table
-- 8. SELECT employee number, employee name WHERE salary is
--      BETWEEN 30000 rs and 50000 rs
-- 9. SELECT employee number, employee name WHERE salary is
--      NOT BETWEEN 30000 and 50000 from the employee table
-- 10. SELECT all the details WHERE city is among the following:
--      Delhi, Mumbai, Chennai, Bangalore
-- 11. SELECT all the details WHERE city is not Mumbai, Delhi
--      or Chennai from the employee table
-- 12. SELECT all the details from the employee table and
--      ORDER BY employee names in descending order
-- 13. Display the list of employees in ascending order from
--      the employee table.
-- =====

DROP TABLE IF EXISTS employee;

-- -----
-- 1(a). Create a table employee with the following attributes:
--      sr no, employee name, job, manager, hire date, salary,
--      department no, city
-- -----
CREATE TABLE employee (
    sr_no          INTEGER PRIMARY KEY,
    employee_name  TEXT NOT NULL,
    job            TEXT NOT NULL,
    manager        TEXT,
    hire_date      DATE NOT NULL,
    salary          NUMERIC(10, 2) NOT NULL CHECK (salary > 0),
    department_no  INTEGER NOT NULL,
    city           TEXT NOT NULL
);

-- -----
-- 1(b). Insert 10 rows in the above mentioned table
--      • salaries below, inside, and above 30000-50000
--      • cities: Delhi, Mumbai, Chennai, Bangalore, Hyderabad, Pune
--      • mixed department numbers (some duplicates)
--      • Rajesh Kumar is the top manager (manager IS NULL)
-- -----
INSERT INTO employee
(sr_no, employee_name, job, manager, hire_date, salary, department_no, city)
VALUES
(1, 'Rajesh Kumar', 'General Manager', NULL, '2018-03-12', 85000.00, 10, 'Delhi'),
(2, 'Ananya Sharma', 'Software Engineer', 'Rajesh Kumar', '2021-07-01', 45000.00, 20, 'Bangalore'),
(3, 'Rohan Mehta', 'Accountant', 'Rajesh Kumar', '2020-01-15', 28000.00, 30, 'Mumbai'),
(4, 'Priya Nair', 'HR Executive', 'Rajesh Kumar', '2022-04-20', 32000.00, 40, 'Chennai'),
(5, 'Vikram Singh', 'Sales Manager', 'Rajesh Kumar', '2019-11-08', 52000.00, 50, 'Hyderabad'),
(6, 'Fatima Khan', 'Software Engineer', 'Ananya Sharma', '2023-02-14', 38000.00, 20, 'Bangalore'),
(7, 'Arjun Patel', 'Clerk', 'Rohan Mehta', '2024-06-01', 18000.00, 30, 'Pune'),
(8, 'Neha Gupta', 'Marketing Analyst', 'Rajesh Kumar', '2021-09-10', 35000.00, 60, 'Delhi'),
(9, 'Karan Iyer', 'Database Admin', 'Ananya Sharma', '2020-12-05', 48000.00, 20, 'Mumbai'),
(10, 'Meera Joshi', 'Receptionist', 'Priya Nair', '2023-08-22', 22000.00, 40, 'Chennai');

-- -----
-- 2. select all the details from the employee table
-- -----
SELECT * FROM employee;

-- -----
-- 3. Select only DISTINCT department number from employee table
--      (lab sheet wrote "distant")
-- -----
```

```

-- Distant (DISTINCT) department number
SELECT DISTINCT department_no FROM employee;

-----
-- 4. select all department numbers from employee table
-----
-- All department numbers
SELECT department_no FROM employee;

-----
-- 5. select employee name, salary where salary is >30000 rs
--    from employee table
-----
-- Salary > 30000 rs
SELECT employee_name, salary
FROM employee
WHERE salary > 30000;

-----
-- 6. select employee name, hire date where city is not delhi
--    from employee table
-----
-- City is not delhi
SELECT employee_name, hire_date
FROM employee
WHERE city <> 'Delhi';

-----
-- 7. select employee name, hire date where city is either
--    delhi or mumbai from employee table
-----
-- City is either delhi or mumbai
SELECT employee_name, hire_date
FROM employee
WHERE city = 'Delhi' OR city = 'Mumbai';

-----
-- 8. select employee number, employee name where salary is
--    between 30000 rs and 50000 rs
-----
-- Salary between 30000 rs and 50000 rs
SELECT sr_no, employee_name
FROM employee
WHERE salary BETWEEN 30000 AND 50000;

-----
-- 9. select employee number, employee name where salary is
--    not between 30000 and 50000 from employee table
-----
-- Salary not between 30000 and 50000
SELECT sr_no, employee_name
FROM employee
WHERE salary NOT BETWEEN 30000 AND 50000;

-----
-- 10. select all the details where city is among the following:
--     delhi, mumbai, chennai, Bangalore
-----
-- City among delhi, mumbai, chennai, Bangalore
SELECT *
FROM employee
WHERE city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore');

-----
-- 11. select all the details where city is not mumbai, delhi
--     or chennai from employee table
-----
-- City is not mumbai, delhi or chennai
SELECT *
FROM employee
WHERE city NOT IN ('Mumbai', 'Delhi', 'Chennai');

-----
-- 12. select all the details from the employee table and
--     order by employee names in descending order.
-----
-- Order by employee names descending
SELECT *
FROM employee
ORDER BY employee_name DESC;

-----
-- 13. display the list of employees in ascending order from
--     employee table.
-----
-- Display employees in ascending order
SELECT *
FROM employee
ORDER BY employee_name ASC;

```

7. Output

The script was executed in SQLite in memory. CREATE and INSERT print a status line; each SELECT prints its result set. NULL in the manager column is shown as an empty cell.

```
SQL> DROP TABLE IF EXISTS employee;
-- OK

SQL> CREATE TABLE employee (
    sr_no      INTEGER PRIMARY KEY,
    employee_name TEXT NOT NULL,
    job        TEXT NOT NULL,
    manager    TEXT,
    hire_date  DATE NOT NULL,
    salary     NUMERIC(10, 2) NOT NULL CHECK (salary > 0),
    department_no INTEGER NOT NULL,
    city      TEXT NOT NULL
);
-- OK

SQL> INSERT INTO employee
(sr_no, employee_name, job, manager, hire_date, salary, department_no, city)
VALUES
(1, 'Rajesh Kumar', 'General Manager', NULL, '2018-03-12', 85000.00,
10, 'Delhi'),
(2, 'Ananya Sharma', 'Software Engineer', 'Rajesh Kumar', '2021-07-01', 45000.00,
20, 'Bangalore'),
(3, 'Rohan Mehta', 'Accountant', 'Rajesh Kumar', '2020-01-15', 28000.00,
30, 'Mumbai'),
(4, 'Priya Nair', 'HR Executive', 'Rajesh Kumar', '2022-04-20', 32000.00,
40, 'Chennai'),
(5, 'Vikram Singh', 'Sales Manager', 'Rajesh Kumar', '2019-11-08', 52000.00,
50, 'Hyderabad'),
(6, 'Fatima Khan', 'Software Engineer', 'Ananya Sharma', '2023-02-14', 38000.00,
20, 'Bangalore'),
(7, 'Arjun Patel', 'Clerk', 'Rohan Mehta', '2024-06-01', 18000.00,
30, 'Pune'),
(8, 'Neha Gupta', 'Marketing Analyst', 'Rajesh Kumar', '2021-09-10', 35000.00,
60, 'Delhi'),
(9, 'Karan Iyer', 'Database Admin', 'Ananya Sharma', '2020-12-05', 48000.00,
20, 'Mumbai'),
(10, 'Meera Joshi', 'Receptionist', 'Priya Nair', '2023-08-22', 22000.00,
40, 'Chennai');
-- 10 row(s) inserted

SQL> SELECT * FROM employee;
sr_no  employee_name  job              manager          hire_date  salary  department_no  city
-----
1      Rajesh Kumar    General Manager
2      Ananya Sharma   Software Engineer Rajesh Kumar      2021-07-01 45000         20      Bangalore
3      Rohan Mehta     Accountant       Rajesh Kumar      2020-01-15 28000         30      Mumbai
4      Priya Nair      HR Executive     Rajesh Kumar      2022-04-20 32000         40      Chennai
5      Vikram Singh    Sales Manager    Rajesh Kumar      2019-11-08 52000         50      Hyderabad
6      Fatima Khan     Software Engineer Ananya Sharma     2023-02-14 38000         20      Bangalore
7      Arjun Patel     Clerk           Rohan Mehta       2024-06-01 18000         30      Pune
8      Neha Gupta      Marketing Analyst Rajesh Kumar      2021-09-10 35000         60      Delhi
9      Karan Iyer      Database Admin   Ananya Sharma     2020-12-05 48000         20      Mumbai
10     Meera Joshi     Receptionist     Priya Nair        2023-08-22 22000         40      Chennai

SQL> SELECT DISTINCT department_no FROM employee;
department_no
-----
10
20
30
40
50
60

SQL> SELECT department_no FROM employee;
department_no
-----
10
20
30
40
50
20
30
60
20
40

SQL> SELECT employee_name, salary
FROM employee
WHERE salary > 30000;
employee_name  salary
-----
Rajesh Kumar   85000
Ananya Sharma  45000
Priya Nair     32000
Vikram Singh   52000
Fatima Khan    38000
Neha Gupta     35000
Karan Iyer     48000

SQL> SELECT employee_name, hire_date
```

```

FROM employee
WHERE city <> 'Delhi';
employee_name hire_date
-----
Ananya Sharma 2021-07-01
Rohan Mehta   2020-01-15
Priya Nair    2022-04-20
Vikram Singh  2019-11-08
Fatima Khan   2023-02-14
Arjun Patel   2024-06-01
Karan Iyer    2020-12-05
Meera Joshi   2023-08-22

```

```

SQL> SELECT employee_name, hire_date
FROM employee
WHERE city = 'Delhi' OR city = 'Mumbai';
employee_name hire_date
-----
Rajesh Kumar  2018-03-12
Rohan Mehta   2020-01-15
Neha Gupta    2021-09-10
Karan Iyer    2020-12-05

```

```

SQL> SELECT sr_no, employee_name
FROM employee
WHERE salary BETWEEN 30000 AND 50000;
sr_no employee_name
-----
2      Ananya Sharma
4      Priya Nair
6      Fatima Khan
8      Neha Gupta
9      Karan Iyer

```

```

SQL> SELECT sr_no, employee_name
FROM employee
WHERE salary NOT BETWEEN 30000 AND 50000;
sr_no employee_name
-----
1      Rajesh Kumar
3      Rohan Mehta
5      Vikram Singh
7      Arjun Patel
10     Meera Joshi

```

```

SQL> SELECT *
FROM employee
WHERE city IN ('Delhi', 'Mumbai', 'Chennai', 'Bangalore');
sr_no employee_name job manager hire_date salary department_no city
-----
1      Rajesh Kumar General Manager 2018-03-12 85000 10 Delhi
2      Ananya Sharma Software Engineer Rajesh Kumar 2021-07-01 45000 20 Bangalore
3      Rohan Mehta Accountant Rajesh Kumar 2020-01-15 28000 30 Mumbai
4      Priya Nair HR Executive Rajesh Kumar 2022-04-20 32000 40 Chennai
6      Fatima Khan Software Engineer Ananya Sharma 2023-02-14 38000 20 Bangalore
8      Neha Gupta Marketing Analyst Rajesh Kumar 2021-09-10 35000 60 Delhi
9      Karan Iyer Database Admin Ananya Sharma 2020-12-05 48000 20 Mumbai
10     Meera Joshi Receptionist Priya Nair 2023-08-22 22000 40 Chennai

```

```

SQL> SELECT *
FROM employee
WHERE city NOT IN ('Mumbai', 'Delhi', 'Chennai');
sr_no employee_name job manager hire_date salary department_no city
-----
2      Ananya Sharma Software Engineer Rajesh Kumar 2021-07-01 45000 20 Bangalore
5      Vikram Singh Sales Manager Rajesh Kumar 2019-11-08 52000 50 Hyderabad
6      Fatima Khan Software Engineer Ananya Sharma 2023-02-14 38000 20 Bangalore
7      Arjun Patel Clerk Rohan Mehta 2024-06-01 18000 30 Pune

```

```

SQL> SELECT *
FROM employee
ORDER BY employee_name DESC;
sr_no employee_name job manager hire_date salary department_no city
-----
5      Vikram Singh Sales Manager Rajesh Kumar 2019-11-08 52000 50 Hyderabad
3      Rohan Mehta Accountant Rajesh Kumar 2020-01-15 28000 30 Mumbai
1      Rajesh Kumar General Manager 2018-03-12 85000 10 Delhi
4      Priya Nair HR Executive Rajesh Kumar 2022-04-20 32000 40 Chennai
8      Neha Gupta Marketing Analyst Rajesh Kumar 2021-09-10 35000 60 Delhi
10     Meera Joshi Receptionist Priya Nair 2023-08-22 22000 40 Chennai
9      Karan Iyer Database Admin Ananya Sharma 2020-12-05 48000 20 Mumbai
6      Fatima Khan Software Engineer Ananya Sharma 2023-02-14 38000 20 Bangalore
7      Arjun Patel Clerk Rohan Mehta 2024-06-01 18000 30 Pune
2      Ananya Sharma Software Engineer Rajesh Kumar 2021-07-01 45000 20 Bangalore

```

```

SQL> SELECT *
FROM employee
ORDER BY employee_name ASC;
sr_no employee_name job manager hire_date salary department_no city
-----
2      Ananya Sharma Software Engineer Rajesh Kumar 2021-07-01 45000 20 Bangalore
7      Arjun Patel Clerk Rohan Mehta 2024-06-01 18000 30 Pune
6      Fatima Khan Software Engineer Ananya Sharma 2023-02-14 38000 20 Bangalore
9      Karan Iyer Database Admin Ananya Sharma 2020-12-05 48000 20 Mumbai
10     Meera Joshi Receptionist Priya Nair 2023-08-22 22000 40 Chennai
8      Neha Gupta Marketing Analyst Rajesh Kumar 2021-09-10 35000 60 Delhi
4      Priya Nair HR Executive Rajesh Kumar 2022-04-20 32000 40 Chennai
1      Rajesh Kumar General Manager 2018-03-12 85000 10 Delhi
3      Rohan Mehta Accountant Rajesh Kumar 2020-01-15 28000 30 Mumbai
5      Vikram Singh Sales Manager Rajesh Kumar 2019-11-08 52000 50 Hyderabad

```

8. Results

Ten employees load successfully. Each lab question returns a non-empty result on this seed data:

Q. No.	Rows	What the result shows
1(a)	—	Table employee created with 8 attributes
1(b)	10	10 rows inserted
2	10	All details from the employee table
3	6	DISTINCT department numbers: 10, 20, 30, 40, 50, 60
4	10	All department numbers (dept 20 appears three times)
5	7	Employee name and salary where salary is > 30000 rs
6	8	Employee name and hire date where city is not Delhi
7	4	Employee name and hire date where city is Delhi or Mumbai
8	5	Employee number and name where salary is between 30000 rs and 50000 rs
9	5	Employee number and name where salary is not between 30000 and 50000
10	8	All details where city is Delhi, Mumbai, Chennai, or Bangalore
11	4	All details where city is not Mumbai, Delhi, or Chennai
12	10	All details ordered by employee names descending
13	10	List of employees in ascending order

Question 3 (DISTINCT) collapses ten department values to six unique numbers; question 4 lists all ten. Questions 8 and 9 partition salaries with no overlap ($5 + 5 = 10$). Question 11 keeps Bangalore, Hyderabad, and Pune — cities that are not Mumbai, Delhi, or Chennai.

9. Conclusion

A single SELECT can project columns, remove duplicates, filter with comparison and set predicates, and sort the result. DISTINCT answers “which unique values exist?” WHERE answers “which rows match this condition?” BETWEEN and IN express ranges and lists clearly. ORDER BY changes presentation only. Together these clauses are the core of everyday retrieval in SQL.